



Beats earphone explodes

Woman reportedly burned as her Beats earphones caught fire mid flight. Apple blames third-party AAA batteries. <http://dgit.in/BeatsBomb>



2,50,000 Redmi 4 sold

Amazon India has confirmed that it sold more than 2,50,000 units of Xiaomi Redmi 4 within 8 minutes of the sale. <http://dgit.in/xiaomi4a>



A closer look at NVIDIA Volta

NVIDIA's next gen GPU architecture is all set to leapfrog all expectations.

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ascal was one of the biggest leaps in gaming performance that NVIDIA had achieved in recent

years. Maxwell was quite the game changer as well, but nowhere close to what Pascal achieved. However, Volta, NVIDIA's next GPU architecture which will succeed Pascal, seems to be all set to clobber Pascal. At the recent GPU Technology Conference in San Jose, NVIDIA CEO Jen-Hsun Huang unveiled the first card to be built using the Volta microarchitecture, the Tesla V100. Going by the Tesla moniker in the name, you should have guessed that this is no gaming graphics card, and you're right. The GeForce branded GTX 2080, GTX 2070 will be coming

out much later. We can expect them in the first half of 2018. Meanwhile, we can take a closer look at Volta.

TESLA V100 : BUILT FOR MACHINE LEARNING AND HIGH PERFORMANCE COMPUTING

Tesla cards are increasingly being deployed by not only Universities that have need for HPC clusters but even up and coming startups operating in

the Artificial Intelligence / Machine Learning domain. It's this very domain that has posed some unique challenges with respect to the kind of compute tasks that Tesla cards have to handle. And to meet the ridiculously growing needs of High Performance Computing, the Tesla V100 has incorporated an entirely new component in the GPU die, Tensor Cores.

You've obviously heard of CUDA cores, NVIDIA's Compute Unified Device Architecture which did away with specialised components to come up with a jack-of-all-trades processing unit. However, while traditional computing tasks have significantly benefited from the GPGPU compute methodology, the needs of AI and ML vary a little. Enough to warrant an entirely new set of Cores. These Tensor Cores will sit alongside the existing CUDA cores and will step in when needed to run GEMM operations. NVIDIA is claiming that the Tesla V100 can train the ResNet-50 deep neural network about 2.4x faster than the previous gen Tesla P100 based off the Pascal architecture. Moreover, the V100 can perform inference with the ResNet-50 deep neural network about 3.7x faster than the P100. As it stands right now, the Tesla V100 can outdo the Tesla P100 by an average factor of at least 1.5x across operations like DGEMM, FFT, Physics, Seismic and STREAM. All of these are GPGPU compute tasks generally undertaken by enterprise GPUs.

PEERING DOWN THE MICROSCOPE - GV100 GPU

As was the case with all previous graphics cards since CUDA was intro-

